

NIVEDH SUNIL

Backend AI Engineer | Backend Engineer

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PROFESSIONAL SUMMARY

Backend AI Engineer with production experience building agentic systems in Python and FastAPI. Built a 32-bit bare-metal operating system in C and a Transformer from scratch in Zig. Currently Backend AI Engineering Intern at FlyRank AI. Seeking Backend AI Engineer and Backend Engineer roles.

EXPERIENCE

Backend AI Engineering Intern — FlyRank AI · Bengaluru

Jul 2026 – Sep 2026

- Owned and delivered 4+ production agentic backend services in Python/FastAPI used by the AI Engineering cohort
- Designed and implemented REST APIs, authentication, job queues, and LLM tool-calling pipelines end-to-end
- Built backend infrastructure for multi-tenant widgets, metering, webhooks and rate limiting supporting concurrent cohort projects

PROJECTS

N-OS — Bare-Metal Operating System — C, x86 · 2025 – Present

Built a complete 32-bit operating system from scratch including virtual memory, preemptive multitasking, FAT32 filesystem, TCP/IP stack, and binary loaders that execute both Windows PE and Linux ELF binaries. Delivered as a bootable ISO running in VirtualBox on 64MB RAM.

<https://github.com/NivedhN160/N-OS>

Nexus — Personal AI Operations Platform — Python, FastAPI, Docker · Aug 2026

Built a unified self-hosted AI platform combining agent execution, content generation, lead capture, social scheduling, and system auditing into a single Docker Compose stack that runs fully local-first.

<https://github.com/NivedhN160/Nexus>

PinnacleRAG-DS — Hybrid RAG System — Python, FastAPI, ChromaDB · Aug 2026

Built a production-style hybrid RAG pipeline with dense + BM25 retrieval, cross-encoder reranking, citation-enforced generation, domain isolation, and evaluation dashboard.

<https://github.com/NivedhN160/PinnacleRAG-DS>

ZigNGPTv2.0 — Transformer from Scratch — Zig · Jun 2026

Implemented a complete offline AI assistant with custom autograd engine and Transformer architecture trained and executed entirely in pure Zig with zero external ML framework dependencies.

<https://github.com/NivedhN160/ZigNGPTv2.0>

LabMate — Medical AI Report Simplifier (Live Product) — Next.js, LLaMA 3 · May 2026 – Present

Built an AI-powered tool that extracts key metrics from medical lab report PDFs and explains them in plain English with lifestyle suggestions. Shipped as an installable PWA, deployed and publicly accessible.

<https://github.com/NivedhN160/Labmate>

TECHNICAL SKILLS

Languages: Python, C, Zig, TypeScript, JavaScript, Java, C++, SQL, NoSQL

Backend: FastAPI, REST APIs, Docker, PostgreSQL, Redis, SQLite, Caching, Microservices

AI / Agents: Agentic AI, Hybrid RAG, Tool-calling, Groq, MCP, Local LLMs

Systems: Bare-metal OS Development, Virtual Memory, TCP/IP Stack, PE & ELF Loaders

Cloud & Tools: AWS (Bedrock, Lambda, DynamoDB, S3), Next.js, Git, Pytest, CI/CD, OAuth

EDUCATION

B.Tech, Computer Science & Engineering — Garden City University

Sep 2024 – Jun 2028

CGPA: 9.5+ (SGPA: 9.55 / 9.55 / 9.82 / 9.13) | Coursework: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, OOP, AI, Data Science

HACKATHONS & CERTIFICATIONS

Hackathons: AWS Build For India · Elastic × AWS UG Bengaluru Hacknight · Browser-Use GTM Edition · MS Ramaiah Ideathon (Top Performer) · Google Hack2Skill · Growth Hackathon

Certifications: Claude 101 & Claude Cowork (Anthropic) · AI Fluency Framework (Anthropic) · Automata & Computability (BITS Pilani) · AI For Everyone (DeepLearning.AI) · Generative AI Mastery Workshop (NxtWave)